

Abstraction

Abstraction involves looking at a problem and filtering out all the unnecessary information. Identifying the essential parts of a problem helps people to figure out a solution.

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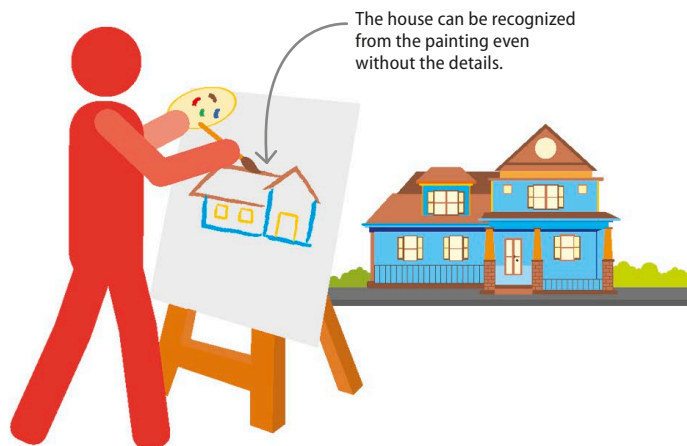
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Bare essentials

Abstraction is the process of working out which elements of an object, or system, are its defining features. Without them the object wouldn't be what it is. Spanish artist Pablo Picasso often painted abstract portraits where faces look nothing like they do in the real world. Yet since they contain the essential features of a face – the eyes, nose, and mouth – they are recognizable.

▷ Which details are unnecessary?

The essential details of a house are the walls, roof, door, and windows. Things like the colour or number of windows are not essential details.



REAL WORLD

Railway maps



Many modern maps of train and subway systems are examples of abstraction. Earlier maps accurately showed the path a railway line took and included the distances between stations. However, they were hard to read as they gave information that passengers didn't need, such as the exact route, and made it difficult to see information that they did need, such as the order of stations on a line. Newer maps simplify the information, so passengers can see the most efficient way to get from A to B.

Finding the right level of detail

The trick with abstraction is to get the balance right between essential and non-essential information. For example, when passing on instructions for a task, too many steps can cause confusion. Too few can mean the task is not done correctly. Computer programmers need to get the balance just right in order for their programs to fix the problem.

The washing machine's controls are an example of abstraction.

▷ Managing complexity

To wash clothes, it is not necessary to understand exactly how a washing machine works. The steps to turn on the machine and program the correct wash are all that is required.

